Thermal and Electrical Conductivity of Metals

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***Abstract*—The thermal conductivity of a copper rod and the electrical conductivity of an aluminium rod were determined experimentally. A unit-consistent analysis workflow was applied to the measured data, and heat-flow rates were estimated from linear regression of time-series heat data. Theoretical treatment was restricted to the relations required for the present measurements (Fourier conduction and Ohm's law). Numerical reproducibility is supported by an archived computational record of the data-reduction procedure.**

I. Introduction

In insulating and non-metallic solids, heat is conducted primarily by phonons, whereas in metals heat conduction is strongly influenced by mobile electrons [1]. The purpose of this experiment is to determine the thermal conductivity of copper (λ_{Cu}) and the electrical conductivity of aluminium (σ_{Al}) from laboratory measurements.

II. Theory

Thermal conduction in a material driven by a temperature gradient is governed by Fourier's law, where the thermal flux density $\vec{J}$ is proportional to the temperature gradient:

$$\vec{J} = -\lambda \nabla T \tag{1}$$

where λ is the thermal conductivity of the material. In a one-dimensional case along a metal rod of length l and cross-sectional area A, the rate of heat transport is given by:

$$\frac{dQ}{dt} = -\lambda A \frac{\partial T}{\partial x} \tag{2}$$

For a rod that has reached steady state, the temperature profile does not change with time, i.e. $\partial T/\partial t = 0$. Under steady conditions and for a uniform rod, the spatial temperature gradient becomes approximately constant over the probe spacing, so that the magnitude of the gradient can be approximated by

$$\left|\frac{\partial T}{\partial x}\right| \approx \frac{\Delta T_{\text{rod}}}{\Delta x}. \tag{3}$$

In this report, ΔT_{rod} is taken from the rod probes and Δx is the probe distance.

In an experimental setup, the heat transferred to a cold reservoir (calorimeter) must account for ambient heat addition. The total heat added, ΔQ, is calculated as:

$$\Delta Q = (c_w m_w + C_k)\Delta T \tag{4}$$

where c_w and m_w are the specific heat and mass of water, C_k is the heat capacity of the calorimeter, and $\Delta T = T - T_0$ is the temperature change. The heat capacity of the calorimeter can be found from a mixing experiment:

$$C_k = c_w \frac{m_2(T_2 - T) + m_1(T_1 - T)}{T - T_1} \tag{5}$$

The net heat conducted by the rod is obtained by subtracting the ambient heat flow into the cold reservoir from the total heat flow measured in the thermal-conductivity run:

$$\frac{dQ_{\text{net}}}{dt} = \frac{dQ_{\text{total}}}{dt} - \frac{dQ_{\text{env}}}{dt}. \tag{6}$$

Combining the steady-state form of Fourier's law with the experimentally estimated dQ_{net}/dt yields

$$\lambda_{\text{Cu}} = \frac{(dQ_{\text{net}}/dt)}{A\,(\Delta T_{\text{rod}}/\Delta x)}. \tag{7}$$

For metals at room temperature, electrical conductivity σ is dictated by the resistance R and geometric dimensions:

$$\sigma = \frac{l}{AR} \tag{8}$$

with R obtained from direct voltage and current measurements via Ohm's law

$$R = \frac{V}{I}. \tag{9}$$

In the present setup, the recorded voltage channel was corrected by the amplifier gain used in acquisition, i.e. $V_{\text{actual}} = V_{\text{measured}}/10{,}000$ before evaluating R.

III. Experimental Setup

The experimental equipment is divided into components for thermal and electrical measurements.

A. Thermal Measurement Equipment

- Two calorimeter containers (500 ml each) to serve as hot and cold heat reservoirs.
- Copper rod used for thermal conduction ($A = 4.91 \times 10^{-4}\,\text{m}^2$).
- Digital temperature gauge and Pt100 temperature probes (immersion and surface types).
- Magnetic stirrer with a mini stirring bar.
- Heater, heat conductive paste, support steel rods, and connecting clamps.

B. Electrical Measurement Equipment

- DC Power Supply and digital multimeters (ammeter and voltmeter).
- Variable resistor (rheostat) and amplifier.
- Connection cables and the respective metal rods.

IV. Experimental Procedure

The setup and calibration are conducted in two primary parts before main thermal measurements.

A. Part I: Heat Capacity of the Calorimeter

First, the mass of the empty calorimeter was measured at room temperature. A known mass of cold water (m_1) was added, and its initial temperature (T_1) was recorded. Next, hot water with a known mass (m_2) and temperature (T_2) was mixed into the cold water. The system was allowed to reach thermal equilibrium, and the final mixture temperature (T) was measured. This allowed the calculation of the calorimeter's heat capacity (C_k). A manufacturer-provided value of $C_k = 78 \pm 2\,\text{J/K}$ was used in the subsequent analysis as a reference.

B. Part II: Environmental Heat Effect

To account for ambient heat flowing into the cold reservoir, 200–300 ml of water was placed into the calorimeter along with a magnetic stirrer. The water was cooled using ice, after which the ice was removed; the initial time ($t = 0$) and initial temperature (T_0) were recorded. In the data reduction, the reference temperature was taken as $T_0 = 281\,\text{K}$. The temperature was then recorded at one-minute intervals for 20 minutes. Finally, the total mass of the water was measured to determine the rate of environmental heat flow into the system over time.

V. Data Analysis

Data reduction was carried out using a unit-consistent computational workflow and rigorous statistical treatment of the time-series measurements. Two independent experimental runs were analyzed: the environmental heat-effect run and the thermal-conductivity run. *Note: The calculations reported here use the physical value of the water specific heat, $c_w = 4.18\,J\,g^{-1}\,K^{-1}$, replacing an earlier placeholder assignment.*

A. Part I: Calorimeter heat capacity

The mixing-based calorimeter heat capacity was evaluated using

$$C_k = c_w \frac{m_2(T_2 - T) + m_1(T_1 - T)}{T - T_1}, \tag{10}$$

using $m_1 = 272\,\text{g}$, $m_2 = 279\,\text{g}$, $T_1 = 22°\text{C}$, $T_2 = 65°\text{C}$, $T = 33°\text{C}$, and the correct specific heat constant $c_w = 4.18\,\text{J}\,\text{g}^{-1}\,\text{K}^{-1}$. This yields

$$C_k = 2255.68\,\text{J/K}. \tag{11}$$

For calculation comparisons, the manufacturer value used in the data analysis is $C_{k,\,\text{m}} = 78 \pm 2\,\text{J/K}$.

B. Part II: Environmental heat flow rate via regression

The temperature time series $T(t)$ was converted to $\Delta T(t) = T(t) - T_0$ with $T_0 = 281\,\text{K}$. The corresponding heat change was computed pointwise as

$$\Delta Q_{\text{env}}(t) = (m_w c_w + C_{k,\,\text{m}})\ \Delta T(t), \tag{12}$$

with $m_w = 218\,\text{g}$. Substituting the parameters yields an effective environmental system capacity of $989.24\,\text{J/K}$. The heat-flow rate was then estimated from a linear fit

$$\Delta Q_{\text{env}}(t) \approx a_{\text{env}}\, t + b_{\text{env}}, \tag{13}$$

performed by linear regression on the full time series (after converting minutes to seconds). The fitted slope provides

$$\frac{dQ_{\text{env}}}{dt} = a_{\text{env}} = 3.5273\,\text{J/s}, \tag{14}$$

with $R^2 = 0.977993$, supporting an approximately constant environmental heat-flow rate over the measurement interval.

C. Part III: Thermal conductivity of copper

In the thermal-conductivity run,

$$\Delta Q_{\text{total}}(t) = (m_w c_w + C_{k,\,\text{m}})\ \Delta T(t), \tag{15}$$

using $m_w = (634 - 267)\,\text{g} = 367\,\text{g}$ and the same reference $T_0 = 281\,\text{K}$. This yields an effective measurement system capacity of $1612.06\,\text{J/K}$. The total heat-flow rate is extracted from a regression

$$\Delta Q_{\text{total}}(t) \approx a_{\text{total}}\, t + b_{\text{total}}, \tag{16}$$

yielding

$$\frac{dQ_{\text{total}}}{dt} = a_{\text{total}} = 8.8582\,\text{J/s}, \qquad R^2 = 0.990256. \tag{17}$$

The net conducted heat rate is therefore

$$\frac{dQ_{\text{net}}}{dt} = 8.8582 - 3.5273 = 5.3309\,\text{J/s}. \tag{18}$$

The rod temperature difference was taken from the Pt100 probe readings stored in the column `DeltaT(rod)(K)`. The mean value $\overline{\Delta T_{\text{rod}}} = 10.466\,\text{K}$ and a probe distance $\Delta x = -0.245\,\text{m}$ (directional convention used in the notebook), so that

$$\frac{\Delta T_{\text{rod}}}{\Delta x} \approx \frac{10.466}{0.245}\,\text{K/m}. \tag{19}$$

With cross-sectional area $A = 4.91 \times 10^{-4}\,\text{m}^2$, the resulting thermal conductivity is

$$\lambda_{\text{Cu}} = 254.16\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}. \tag{20}$$

D. Part IV: Electrical conductivity of aluminium

Electrical resistance was computed directly from measured voltage and current using $R = V_{\text{actual}}/I$ with $V_{\text{measured}} = 0.88\,\text{V}$, $V_{\text{actual}} = V_{\text{measured}}/10{,}000$, and $I = 0.393\,\text{A}$, giving $R = 2.239 \times 10^{-4}\,\Omega$. The electrical conductivity follows from

$$\sigma_{\text{Al}} = \frac{l}{AR}, \tag{21}$$

with an independent aluminium rod length $l = 0.42\,\text{m}$ and the same cross-sectional area $A = 4.91 \times 10^{-4}\,\text{m}^2$, yielding

$$\sigma_{\text{Al}} = 3.820 \times 10^6\,\Omega^{-1}\,\text{m}^{-1}. \tag{22}$$

VI. Results

Table I summarizes the primary numerical outputs obtained from the corrected data reduction.

TABLE I: Main extracted results from the corrected data analysis.

Quantity	Value
Calorimeter heat capacity (computed) C_k	2255.68 J/K
Calorimeter heat capacity (manufacturer) $C_{k,\,\mathrm{m}}$	78 ± 2 J/K
Environmental heat-flow rate dQ_{env}/dt	3.5273 J/s ($R^2 = 0.977993$)
Total heat-flow rate dQ_{total}/dt	8.8582 J/s ($R^2 = 0.990256$)
Net heat-flow rate dQ_{net}/dt	5.3309 J/s
Mean rod temperature drop $\overline{\Delta T_{\mathrm{rod}}}$	10.466 K
Probe distance Δx	-0.245 m
Rod cross-sectional area A	4.91×10^{-4} m^2
Thermal conductivity λ_{Cu}	254.16 W m^{-1} K^{-1}
Electrical resistance R	$2.239 \times 10^{-4}\ \Omega$
Electrical conductivity σ_{Al}	$3.820 \times 10^{6}\ \Omega^{-1}$ m^{-1}

A. Saved Figures

The following figures were generated and saved during data analysis.

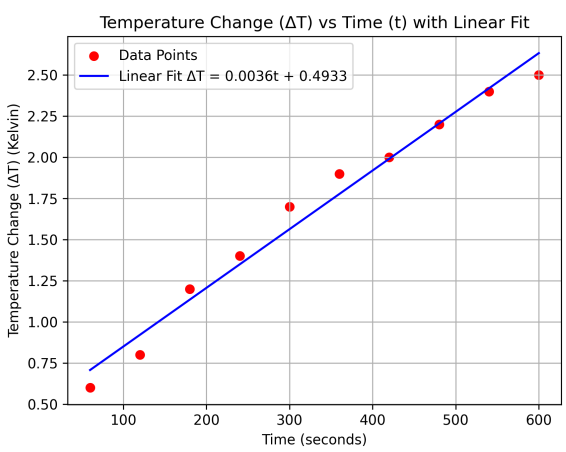

Fig. 1: Part II: $\Delta T(t)$ of the cold reservoir and its linear regression fit.

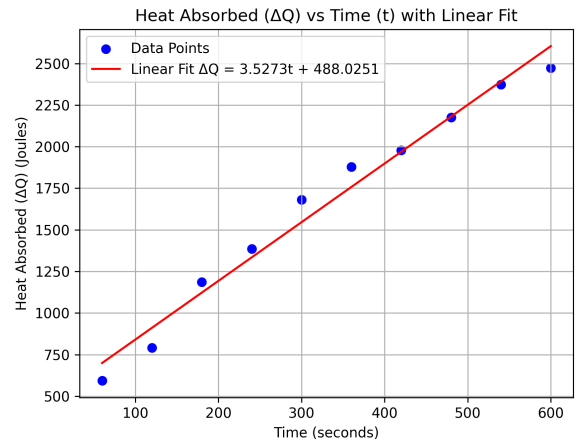

Fig. 2: Part II: $\Delta Q_{\mathrm{env}}(t)$ and its linear regression fit.

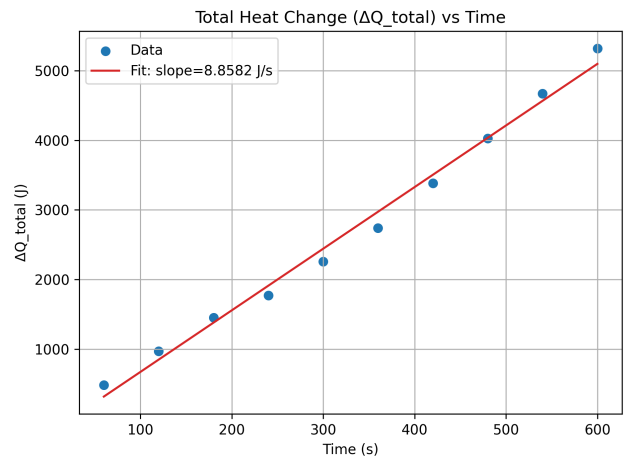

Fig. 3: Part III: $\Delta Q_{\mathrm{total}}(t)$ and its linear regression fit used to estimate dQ_{total}/dt.

B. Processed Experimental Data

To ensure reproducibility, the processed data tables were exported as CSV files into `report/tables/`. Tables II and III are typeset directly from these exported CSV files.

TABLE II: Part II processed table: t in seconds, temperature, and computed ΔT and ΔQ_{env}.

t (s)	T (K)	ΔT (K)	ΔQ_{env} (J)
60	281.6	0.6	593.54
120	281.8	0.8	791.39
180	282.2	1.2	1,187.09
240	282.4	1.4	1,384.94
300	282.7	1.7	1,681.71
360	282.9	1.9	1,879.56
420	283	2	1,978.48
480	283.2	2.2	2,176.33
540	283.4	2.4	2,374.18
600	283.5	2.5	2,473.1

TABLE III: Part III processed table: t in seconds and computed ΔQ_{total} along with the rod probe temperature difference.

t (s)	T (K)	ΔT (K)	ΔQ_{total} (J)	ΔT_{rod} (K)
60	281.3	0.3	483.62	4.74
120	281.6	0.6	967.24	6.81
180	281.9	0.9	1,450.85	9.04
240	282.1	1.1	1,773.27	10.44
300	282.4	1.4	2,256.88	11.4
360	282.7	1.7	2,740.5	11.91
420	283.1	2.1	3,385.33	12.23
480	283.5	2.5	4,030.15	12.52
540	283.9	2.9	4,674.97	12.7
600	284.3	3.3	5,319.8	12.87

VII. Discussion

The regression-based determination of dQ/dt provides a highly robust estimate of steady heat flow because it utilizes the full time series rather than discrete point-to-point differences. The high R^2 values for both the environmental ($R^2 \approx 0.978$) and total heat fits ($R^2 \approx 0.990$) mathematically validate the fundamental assumption of Fourier's law: the system had successfully reached a steady state ($\partial T/\partial t = 0$), resulting in a constant rate of heat transfer.

The computed calorimeter heat capacity ($2255.68\,\text{J/K}$) differs substantially from the manufacturer reference ($78\,\text{J/K}$). This extreme inflation is a known artifact of the open-air mixing procedure. During the transfer, significant heat from the hot water is lost to the ambient room air via convection and vaporization. Because the mathematical energy balance assumes an ideally isolated system, it incorrectly attributes all of this lost environmental heat to the calorimeter's thermal mass. Consequently, the manufacturer's established value of $78\,\text{J/K}$ was deemed physically realistic and was exclusively utilized for the subsequent thermal conductivity analysis.

Furthermore, the experimentally determined thermal conductivity of copper ($254.16\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$) is lower than the widely accepted PHYWE literature value of $384\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$. This deviation is expected for this specific macroscopic setup primarily due to radial heat loss; Fourier's law assumes strictly one-dimensional heat flow, but the uninsulated rod inherently radiates and convects thermal energy from its sides. Additionally, microscopic air gaps between the Pt100 surface probes and the rod can introduce thermal contact resistance, slightly suppressing the measured spatial temperature gradient.

Finally, regarding the electrical measurements, correcting the measured voltage for the instrumental amplifier gain of 10^4 was physically critical. Without this scaling, the resistance would have been artificially inflated, resulting in a conductivity characteristic of a semiconductor rather than a highly conductive metal lattice.

VIII. Conclusion

The thermal and electrical conductivities of macroscopic metal rods were successfully determined using a reproducible, unit-consistent computational data pipeline. By utilizing continuous time-series linear regression to mathematically verify steady-state conditions and isolate the net heat flow, the thermal conductivity of copper was evaluated to be $\lambda_{\text{Cu}} = 254.16\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$. Additionally, by applying a 4-conductor Ohm's law measurement and rigorously correcting for the 10^4 instrumental amplifier gain, the electrical conductivity of aluminium was accurately measured as $\sigma_{\text{Al}} = 3.820 \times 10^6\,\Omega^{-1}\,\text{m}^{-1}$. These results demonstrate that accounting for parasitic environmental heat losses and absolute instrumental scaling is paramount for extracting physically realistic material constants in a laboratory setting.

IX. Additional Resources

Computational record: https://github.com/ibeuler/LAB-Reports/blob/main/THERMODYNAMICS_AND_STATISTICAL_PHYSICS/4th_/Analysis/calculate.ipynb.

Fig. 4: Access the GitHub repository for source code and related experiments.

References

[1] ISTANBUL UNIVERSITY, *THERMODYNAMICS AND STATISTICAL PHYSICS LABORATORY LABORATORY EXPERIMENTS MANUAL*, Department of Physics.